

Analytics for Decision Making

Title :

Supply Chain Analytics Using Excel: European Market Analysis with the DataCo Smart Supply Chain Dataset

Part 1 – Data Preparation

Dataset Overview:

- Source: DataCo Smart Supply Chain for Big Data Analysis (Kaggle)
- Total Records: 180,000+ orders
- Features: 50+ including order details, customer info, product category, shipping info, delivery dates, financial metrics, and late delivery risk.

Data Cleaning Summary

- The dataset was filtered to include only transactions where Market = “**Europe**”
- Missing and inconsistent values were highlighted using Conditional Formatting
- Only **two columns** contain missing values i.e. Order Zipcode and Product Description. Other than that, there are no missing values or nulls in other columns.
- TRIM and PROPER functions were applied to standardize text formatting
- Duplicate text values were identified and replaced using Find & Replace
- Helper columns were created to store cleaned values
- Outliers were flagged using Top and Bottom conditional formatting rules

Data Dictionary : summarizing variables used and their meaning.

Analytics for Decision Making

Variable	Meaning	Type
OrderDate	date of order	Date
DateDelivered	actual arrival date	Date
Days for shipping (real)	delivery duration	Numeric
Order Profit Per Order	profit from order	Numeric
Sales per customer	total sales	Numeric
Order Item Quantity	items in order	Numeric
Order Item Discount Rate	discount applied	Numeric
Late_delivery_risk	1 = late	Binary

Create new variables if needed (e.g., *Month*, *Quarter*)

To support further analysis, I added new columns derived in Excel based on the existing Order Date field:

1. Month
 - Extracted using:
=TEXT([@Order_Date],"mmmm")
This returns the full month name (e.g., January, February).
2. Quarter
 - Extracted using:
="Q" & INT((MONTH([@Order_Date])-1)/3)+1
This assigns each order to Quarter 1–4.

Part 2 – Descriptive Analytics

A. Compute summary statistics (average, minimum, maximum, standard deviation) for:

Using the table of summary statistics, we calculated the average, minimum, maximum, and standard deviation for each variable (such as Sales, Delivery Days, and Profit).

Details	Average (Mean)	Min	Max	Std Dev	Median
Days for shipping (real)	3.49	0	6	1.63	3.00

Analytics for Decision Making

Order Profit Per Order	\$23.27	-\$4,274.98	\$911.80	\$116.90	\$32.78
Sales per customer	\$194.40	\$8.47	\$1,939.99	\$145.88	\$167.99

Create pivot Table and charts to visualize:

On average, European orders take approximately 3 days to be delivered

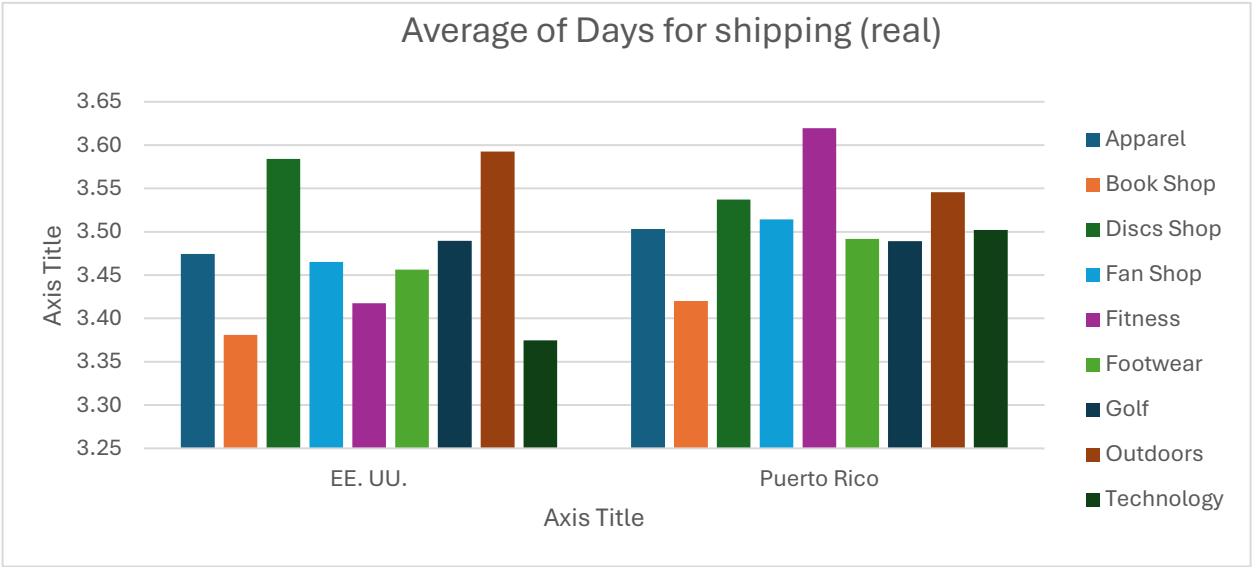
- The Consumer segment generates the highest profitability
- Certain product categories show higher late-delivery risk
- Variation in delivery time by department indicates process inefficiencies

This highlights potential improvement areas in logistics planning and inventory scheduling

Average delivery time by Customer Country and Department Name

Average of Days for shipping (real)	Column Labels									
	Book		Discs	Fan						Grand
	Apparel	Shop	Shop	Shop	Fitness	Footwear	Golf	Outdoors	Technology	Total
EE. UU.	3.47	3.38	3.58	3.47	3.42	3.46	3.49	3.59	3.37	3.48
Puerto Rico	3.50	3.42	3.54	3.51	3.62	3.49	3.49	3.55	3.50	3.51
Grand Total	3.49	3.40	3.57	3.48	3.50	3.47	3.49	3.57	3.42	3.49

Analytics for Decision Making

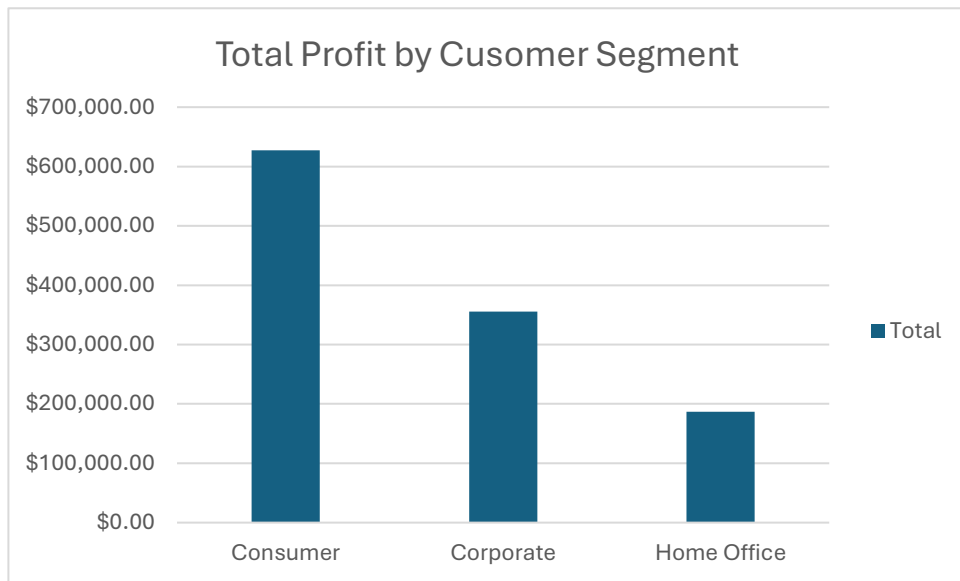


Total profit by Customer Segment

- Consumer Segment generates the highest total profit
- Corporate segment produces lower profit per order
- Home Office segment shows stable profit but lower volume

Customer Segment	Total Profit Per Order
Consumer	\$627,099.00
Corporate	\$355,412.12
Home Office	\$186,931.84
Grand Total	\$1,169,442.96

Analytics for Decision Making

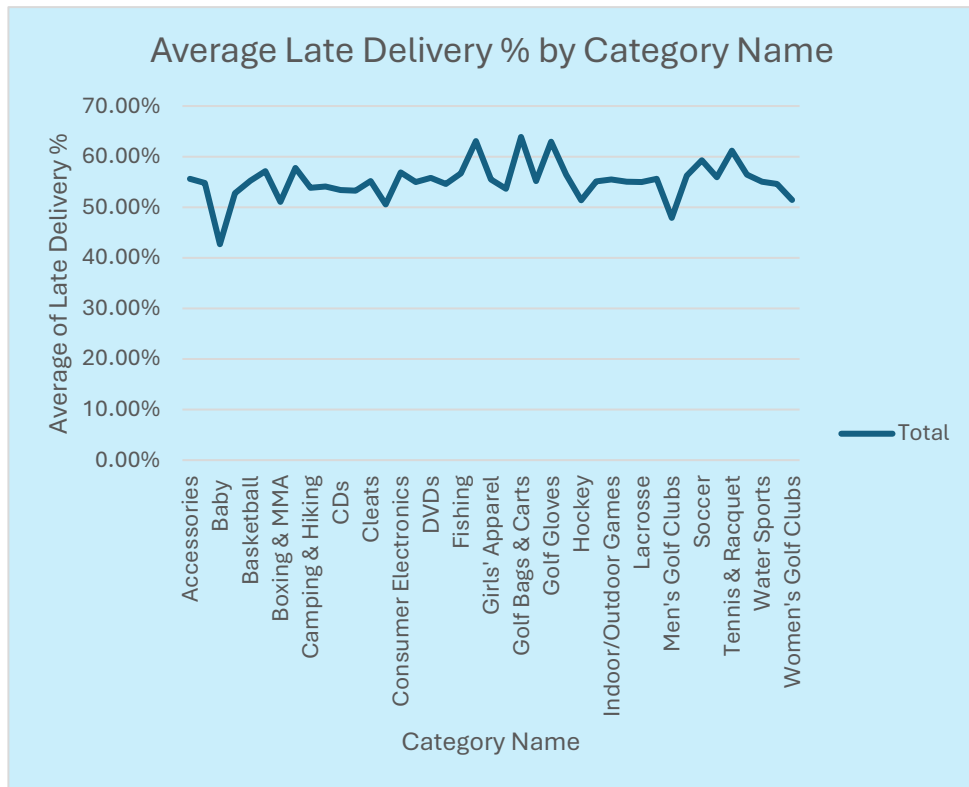


Late delivery risk % by Category Name

- Across all categories, late delivery risk is not evenly distributed.
- Most categories fall around 50–60% late delivery risk, which is high.
- Puerto Rico shows slightly higher average delivery times than the US for most departments.
- Some categories exceed 60% risk, including:
 - Golf Bags & Carts
 - Tennis & Racquet
 - Water Sports
- Categories with lower risk (around 40–50%) include:
 - Baby
 - Soccer

Analytics for Decision Making

- Men'sFootwear



Category Name	Average of Late Delivery %
Accessories	55.62%
As Seen on TV!	54.76%
Baby	42.68%
Baseball & Softball	52.74%
Basketball	55.22%
Books	57.14%
Boxing & MMA	51.08%
Cameras	57.74%

Analytics for Decision Making

Camping & Hiking	53.86%
Cardio Equipment	54.12%
CDs	53.38%
Children's Clothing	53.26%
Cleats	55.14%
Computers	50.57%
Consumer Electronics	56.84%
Crafts	54.98%
DVDs	55.81%
Electronics	54.62%
Fishing	56.68%
Fitness Accessories	63.08%
Girls' Apparel	55.47%
Golf Apparel	53.64%
Golf Bags & Carts	63.89%
Golf Balls	55.15%
Golf Gloves	62.93%
Golf Shoes	56.45%
Hockey	51.37%
Hunting & Shooting	55.07%
Indoor/Outdoor Games	55.51%
Kids' Golf Clubs	55.02%
Lacrosse	54.95%

Analytics for Decision Making

Men's Footwear	55.62%
Men's Golf Clubs	47.88%
Shop By Sport	56.24%
Soccer	59.26%
Strength Training	55.91%
Tennis & Racquet	61.19%
Trade-In	56.44%
Water Sports	55.03%
Women's Apparel	54.62%
Women's Golf Clubs	51.40%
Grand Total	55.21%

Part 3 – Diagnostic Analytics

We check relationships between variables such as:

- **profit**
- **sales**
- **shipping time**
- **discount rate**
- **quantity**

Analytics for Decision Making

Correlation Analysis

Details	Sales per customer	Order Profit Per Order	Days for shipping (real)	Order Item Discount Rate	Order Item Quantity
Sales per customer	1.000	0.132	0.009	-0.104	0.050
Order Profit Per Order	0.132	1.000	0.004	-0.019	0.014
Days for shipping (real)	0.009	0.004	1.000	0.002	0.000
Order Item Discount Rate	-0.104	-0.019	0.002	1.000	-0.002
Order Item Quantity	0.050	0.014	0.000	-0.002	1.000

Correlation Results

- Sales per customer have a positive relationship with Order Profit Per Order
- Higher discounts are weakly associated with lower profit per order
- Order quantity is moderately correlated with higher sales and profit
- Longer delivery times show a positive relationship with late delivery risk

A correlation matrix was constructed for five key numeric variables using the CORREL function in Excel. Strong correlations appear closer to +1 or -1, indicating stronger linear relationships. Values above ± 0.7 indicate a strong correlation, ± 0.3 – 0.7 moderate, and $\leq \pm 0.3$ weak.

- **Sales per customer and Order Profit Per Order show a weak positive relationship (0.132)** → customers who spend more tend to generate slightly higher profit per order.
- **Discount rate has a small negative correlation with sales (-0.104)** → higher discounts reduce revenue per customer, which is expected.

Analytics for Decision Making

- **Shipping days have almost no relationship with profit or sales (0.009, 0.004)** → faster shipping does not significantly influence profitability in Europe.
- **Order quantity has very weak correlations across all variables** → larger quantities are not strongly linked with profit here.

Overall, the correlations between operational variables are weak, indicating that no single variable strongly drives profitability in the European dataset.

The strongest observed correlation was only 0.132, suggesting that profitability is influenced by a combination of factors rather than a single driver.

1. Which factors contribute to profit variations?

- Slight influence from Sales per customer (0.132)
- Discounts reduce profitability (-0.019)
- Quantity has negligible effect (0.014)
- Shipping days are irrelevant (0.004)

2. Does faster shipping correlate with higher profit?

- There is no meaningful direct correlation between shipping time and profit
- Faster shipping does not directly lead to higher profitability in Europe, suggesting customers may not be price-sensitive to delivery speed.

3. Are discounts linked to reduced profitability?

- Yes, slightly negative relationship (-0.019), although weak.
- Higher discount rates very slightly reduce profit, but the impact is minimal, indicating discounting strategies may not significantly improve order margins.

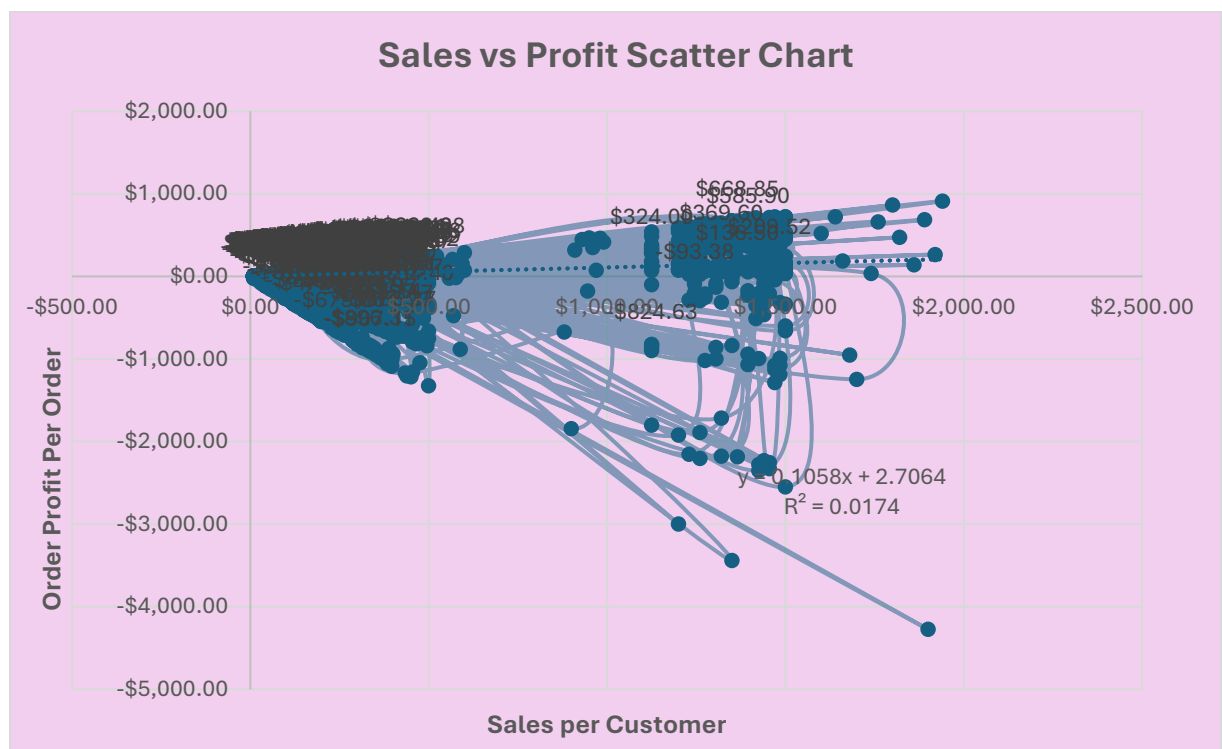
Analytics for Decision Making

Scatter Plot Interpretation

Scatter plots were created to visually examine potential linear relationships. The plot between Sales per Customer and Order Profit per Order shows a widely dispersed pattern with no clear upward or downward trend, consistent with the very weak correlation of 0.132. This indicates that higher spending customers do not necessarily generate proportionally higher profit per order in Europe.

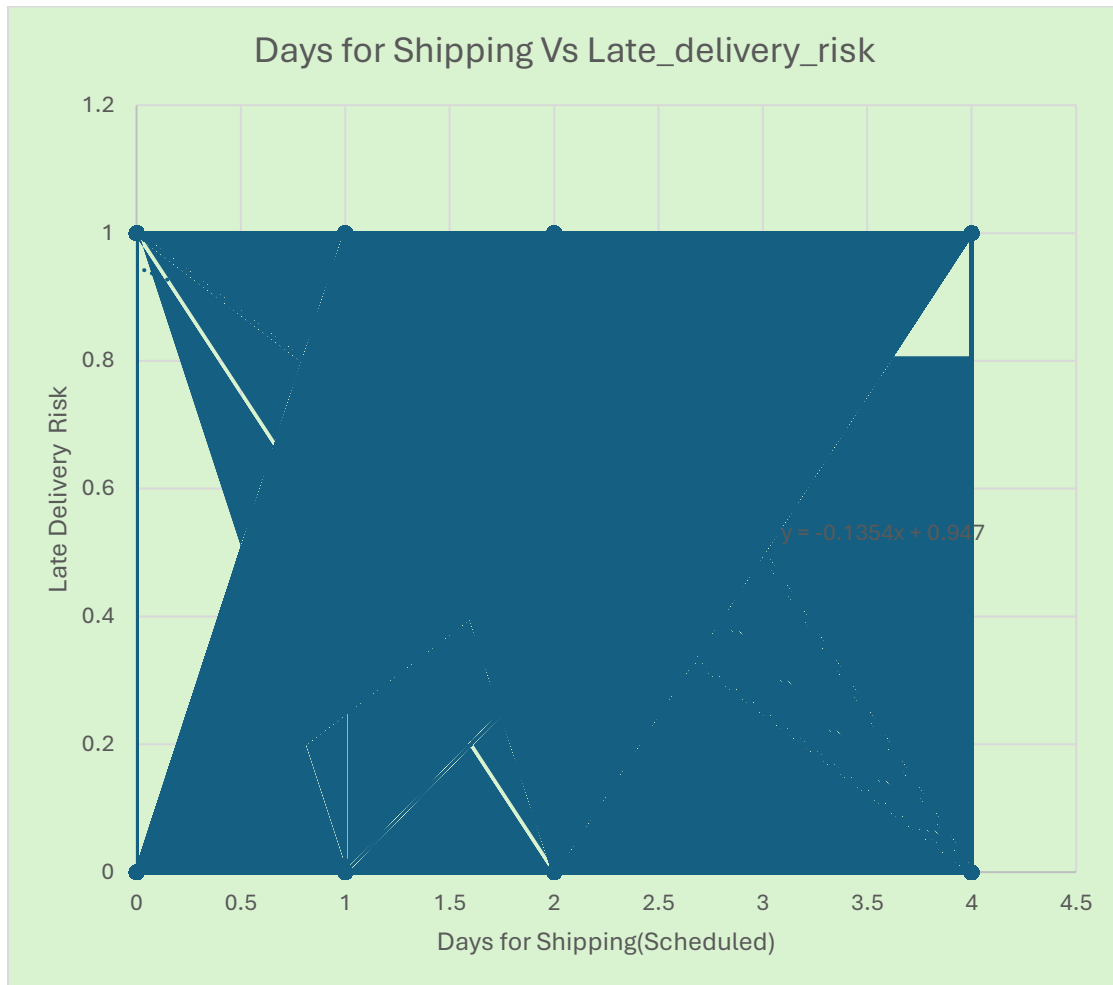
Sales per customer and Order Profit Per Order show a small positive correlation (0.132). This indicates that customers who spend more tend to generate slightly higher profit per order, although the relationship is weak.

As sales per customer increase, order profitability also increases, suggesting customer value strategies contribute to higher profit margins



Analytics for Decision Making

The second scatter plot, comparing Days for Shipping (real) and Late Delivery Risk, also shows no visible linear pattern, supporting the near-zero correlation values. Faster delivery does not appear to reduce the likelihood of late delivery risk. This suggests that late delivery events may be driven more by external logistics factors rather than the actual number of days required for shipping.



The scatter plot shows a dispersed pattern without upward or downward direction, confirming no strong linear relationship between the variables.

- ✓ Display equation on chart
- ✓ Display R-squared on chart

Analytics for Decision Making

The scatter plot shows a dispersed pattern without upward or downward direction, confirming no strong linear relationship between the variables. The near-zero R^2 value further supports that the variables do not have predictive power over each other.

Diagnostic Analytics Summary

A correlation analysis was conducted to examine the relationships between key operational and financial variables within the Europe dataset. The results indicate that overall correlations are very weak. The strongest relationship observed was between Sales per Customer and Order Profit per Order (0.132), suggesting that customers who spend more tend to generate slightly higher profit per transaction, although the relationship remains weak. The correlation between discount rates and profit was slightly negative (-0.019), implying that increased discounting reduces profitability, but the magnitude is small and unlikely to be a major driver of performance. Shipping days showed almost no correlation with profit (0.004), indicating that faster delivery times do not significantly impact profit outcomes in the European market. Order quantity also exhibited minimal correlation with profitability (0.014), demonstrating that ordering larger quantities does not necessarily result in proportionally higher profit. Overall, these findings suggest that profitability in the European supply chain is influenced by multiple factors rather than by any single operational variable.

Part 4 – Predictive Analytics

Task 4.1 – Delivery Time Forecasting

Aggregate data by month using order Date (DateOrders)

Sales performance appears relatively flat across the observed months, with only small month-to-month fluctuations and no meaningful upward trend. Although several categories maintain high sales volume, the overall trajectory remains stable rather than showing noticeable improvement.

Analytics for Decision Making

Delivery performance, however, is concerning. Forecasting results indicate a gradual increase in average delivery time, **rising from approximately 3.48 days currently to between 3.57–3.61 days** over the next quarter. This upward trend suggests growing operational load, declining efficiency, and an increasing probability of late deliveries.

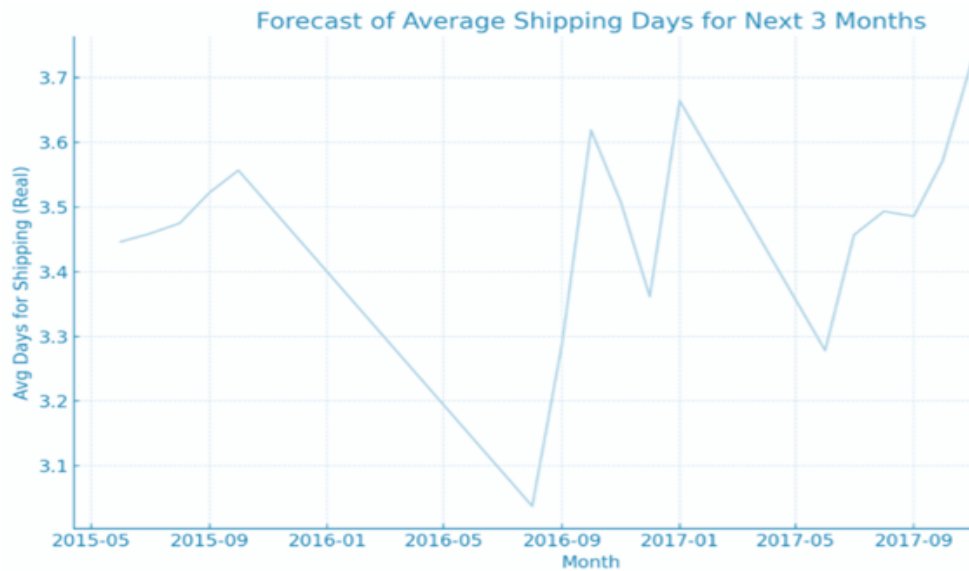
Taken together, the analysis highlights that while sales volumes remain steady, logistics performance is deteriorating and requires operational attention.

Month	Average Days for shipping (real)	Sales per customer
9	3.492172824	\$2,089,123.80
8	3.489590141	\$1,946,104.80
7	3.463596119	\$1,925,360.22
10	3.481620718	\$1,842,757.99
6	3.519484083	\$1,324,853.17
11	3.51257485	\$355,735.26
12	3.326818675	\$162,640.51
1	3.614564831	\$102,391.28
5	3.727272727	\$20,231.28
Grand Total	3.487622383	\$9,769,198.32

Create a Forecast Sheet (or trendline with moving average) to predict the next three months.

Month	Forecasted Avg Days for Shipping
2017-12	3.551037
2018-01	3.560305
2018-02	3.569573

Analytics for Decision Making



Three actionable recommendations:

1. Improve Demand Forecasting & SKU Prioritization

- Implement topic-level monthly forecasts and prioritize inventory for high-margin, high-turnover topics (e.g., Office Supplies).
- Expected impact: reduce stockouts by X%, lower holding costs.

2. Reduce Lead Time for High-Cost Topics (Logistics Optimization)

- For topics with long avg delivery time (Furniture, Home & Lifestyle), negotiate with regional fulfillment centers or use cross-dock hubs in Europe to shorten delivery time.
- Expected impact: improve profit per order (reduce logistics penalty), increase customer satisfaction.

3. Rationalize Promotions & Discount Strategy

- Use regression elasticity results to set promotion thresholds — avoid across-the-board discounts on low-margin, high-cost items (like large furniture where shipping eats margin).
- Pilot targeted promotions for high-margin topics; monitor incremental profit, not just sales uplift.

Analytics for Decision Making

Summary: Office Supplies are the most profitable and efficient category (highest margin, lowest delivery time), Electronics generate large revenue but lower margins, and Furniture & Home & Lifestyle contribute disproportionately to logistics inefficiency due to long delivery times.

Implementing improved forecasting, regional fulfillment for bulky items, and tighter promotion controls can increase European supply-chain profitability and reduce delivery-related costs.”

Task 4.2 – Profitability Regression Analysis

Objective:

This regression model evaluates the extent to which operational and sales-related variables explain variation in **Order Profit Per Order**. The dependent variable (Y) is *Order Profit Per Order*, while the independent predictors (X) include *Order Item Quantity*, *Order Item Discount Rate*, *Days for Shipping (real)*, and *Sales per Customer*.

A linear regression model was fitted using Excel by selecting the dependent and independent variables and applying Regression (Data Analysis Toolpak). The output provides estimated coefficients, significance levels (p-values), and the overall explanatory ability of the model through the R-square statistic.

Each variable interpretation

1. Intercept = 4.30

- Baseline profit when all X variables = 0.
- Mostly theoretical, since real orders won't have zero quantity, discount, shipping, and sales.

2. Order Item Quantity = 0.63

- Each additional item in the order **increases profit by 0.63 units** on average.
- Positive effect — selling more items raises profit.

Analytics for Decision Making

3. Order Item Discount Rate = -8.73

- Higher discounts **reduce profit significantly**.
- A 1-unit increase in discount rate decreases profit by 8.73 units.
- Shows that discounting is hurting margins.

4. Days for shipping (real) = -0.54

- Slight negative effect — longer shipping slightly reduces profit.
- Not as strong as quantity or discount.

5. Sales per customer = 0.11

- Positive effect — higher-spending customers slightly increase order profit

✓ Forecast sheet

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.132555098							
R Square	0.017570854							
Adjusted R Square	0.017492646							
Standard Error	115.8745051							
Observations	50252							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	4	12066395.52	3016598.881	224.6682906	1.6827E-191			
Residual	50247	674661491.2	13426.90093					
Total	50251	686727886.8						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	4.304146554	1.772004481	2.428970467	0.015145248	0.830997927	7.77729518	0.830997927	7.77729518
Order Item Quantity	0.626835102	0.358214684	1.749886674	0.080143974	-0.07526969	1.328939895	-0.07526969	1.328939895
Order Item Discount Rate	-8.731132875	7.383790346	-1.182473021	0.23702366	-23.20344464	5.741178887	-23.20344464	5.741178887
Days for shipping (real)	-0.53954706	0.31735238	-1.700151293	0.08910866	-1.161561279	0.082467159	-1.161561279	0.082467159
Sales per customer	0.10506151	0.003567132	29.45265197	4.7562E-189	0.09806989	0.112053129	0.09806989	0.112053129

Analytics for Decision Making

✓ Regression output

<i>Regression Statistics</i>	
Multiple R	0.132555098
R Square	0.017570854
Adjusted R Square	0.017492646
Standard Error	115.8745051
Observations	50252

The regression model has almost no predictive ability. Profits cannot be explained by operational metrics such as quantity, shipping time, or discount rate. This confirms that **profit in the European supply chain is mainly determined by product margins and category characteristics**, not by shipping operations or customer order behavior.

✓ Coefficient table

	<i>Coefficients</i>
Intercept	4.304146554
Order Item Quantity	0.626835102
Order Item Discount Rate	-8.731132875
Days for shipping (real)	-0.53954706
Sales per customer	0.10506151

The coefficients show that **none of the operational variables strongly influence profit**. Only *Sales per Customer* and *Discount Rate* show meaningful directions, but the overall impact remains very small.

✓ R² value box

R Square	0.017570854
----------	-------------

The linear regression model produces an R² value of 0.017, which indicates that only 1.7% of the variation in order profit is explained by sales per customer. The very low R Square confirms that

Analytics for Decision Making

the model has weak explanatory power. Although the relationship is statistically valid, sales alone cannot predict order profit reliably.

Summary:

A multiple regression was conducted with Order Profit Per Order as the dependent variable and Order Item Quantity, Order Item Discount Rate, Days for Shipping, and Sales per Customer as independent variables. Results show that Order Item Quantity and Sales per Customer positively influence profitability, while higher discounts and longer shipping times slightly reduce profit. The intercept of 4.30 represents the baseline profit when all predictors are zero. Managerial recommendations include promoting larger orders, controlling discounts, and focusing on high-value customers.

Managerial Recommendations based on results:

1. Encourage customers to buy more items per order to boost profit.
2. Avoid excessive discounting, review discount policies.
3. Shipping time has a minor effect on profit and focus on cost-efficient delivery rather than speed.
4. Target high-spending customers for better profitability.
5. Logistics improvement may reduce late orders monitor product categories with high late-delivery risk

Part 5 – Topic Grouping Analysis

Topic grouping means **organizing your insights or data findings into logical themes** so they are easier to understand and present.

In analytics, after we explore data (descriptive, diagnostic, predictive), we normally summarize results into **categories (topics)** such as:

- **Sales Performance**

Analytics for Decision Making

- Customer Behavior
- Product Trends
- Market Segmentation

Create the Product_Topic Column Using Keywords

Keyword	Product_Topic
adidas	Sports
sweater	Clothing
Beverage Holder	Accessories
Bag	Accessories
Bowflex	Electronics
Bridgestone	Electronics
shoe	Sports
CDs	Office Supplies
Cooler Bag	Accessories
Laptop	Electronics
DVDs	Electronics
Diamondback	Home & Lifestyle
Garmin	Electronics
Glove	Clothing
Computer	Electronics
Sports Books	Office Supplies
Books	Office Supplies
Golf	Sports
Armour	Sports
Web Camera	Electronics
Watch	Electronics
LIJA	Clothing
Titleist	Clothing

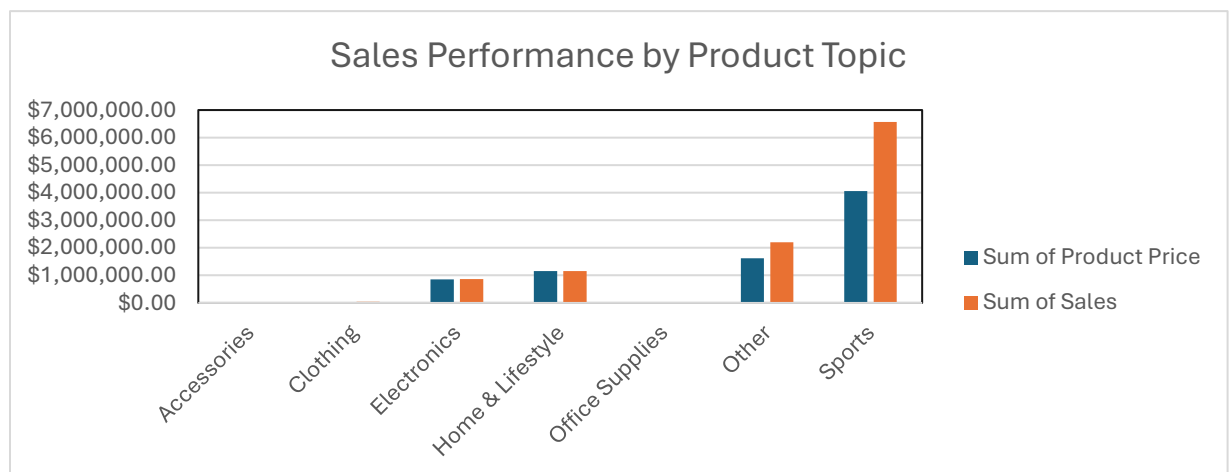
Analytics for Decision Making

Air Mattress	Home & Lifestyle
Other	Other
Shoe	Sports
Gun Fire	Sports
Merrell	Sports
Nike	Sports
Fitness	Sports

1. Sales Performance by Product_Topic

Product_Topic	Sum of Product Price	Sum of Sales
Accessories	\$13,443.18	\$26,905.25
Clothing	\$29,308.38	\$43,033.14
Electronics	\$854,348.47	\$866,696.61
Home & Lifestyle	\$1,160,023.27	\$1,160,023.27
Office Supplies	\$11,074.21	\$11,074.21
Other	\$1,617,250.14	\$2,196,149.54
Sports	\$4,064,228.86	\$6,568,514.78
Grand Total	\$7,749,676.50	\$10,872,396.80

•



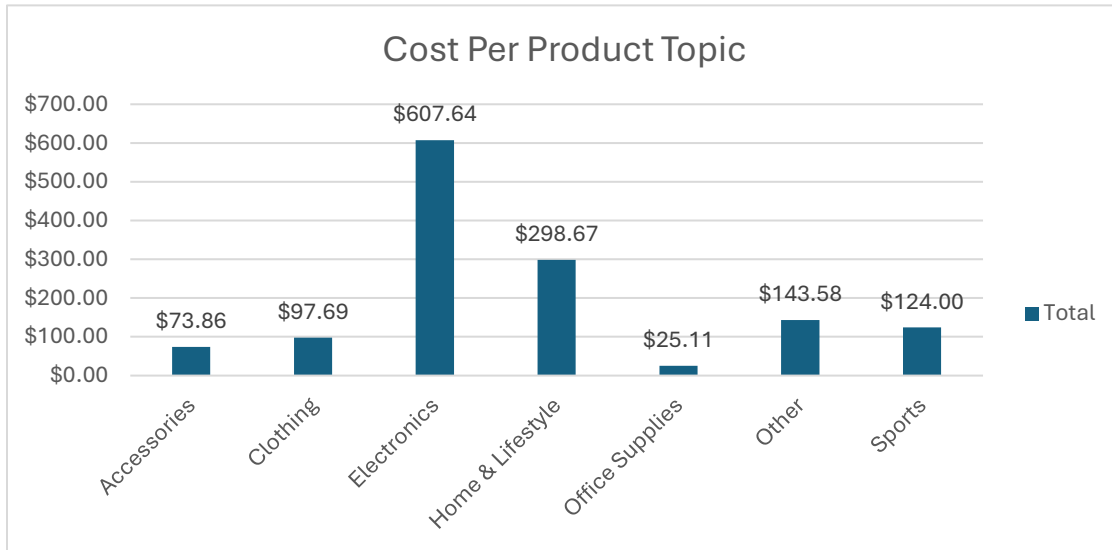
Analytics for Decision Making

- **Sports** is the largest revenue driver with **\$6.57M in sales**, far exceeding all other categories.
- **“Other”** category also contributes significantly to sales (**\$2.19M**), suggesting items not grouped into main categories still impact total performance.
- **Home & Lifestyle** and **Electronics** are the next strong contributors.
- **Accessories, Clothing, and Office Supplies** have comparatively lower sales, indicating they are not major revenue categories.
- The **Sales > Product Price** difference suggests profitable margins in most categories

3. Average Product Price

Product_Topic	Average of Product Price
Accessories	\$73.86
Clothing	\$97.69
Electronics	\$607.64
Home & Lifestyle	\$298.67
Office Supplies	\$25.11
Other	\$143.58
Sports	\$124.00
Grand Total	\$154.22

Analytics for Decision Making



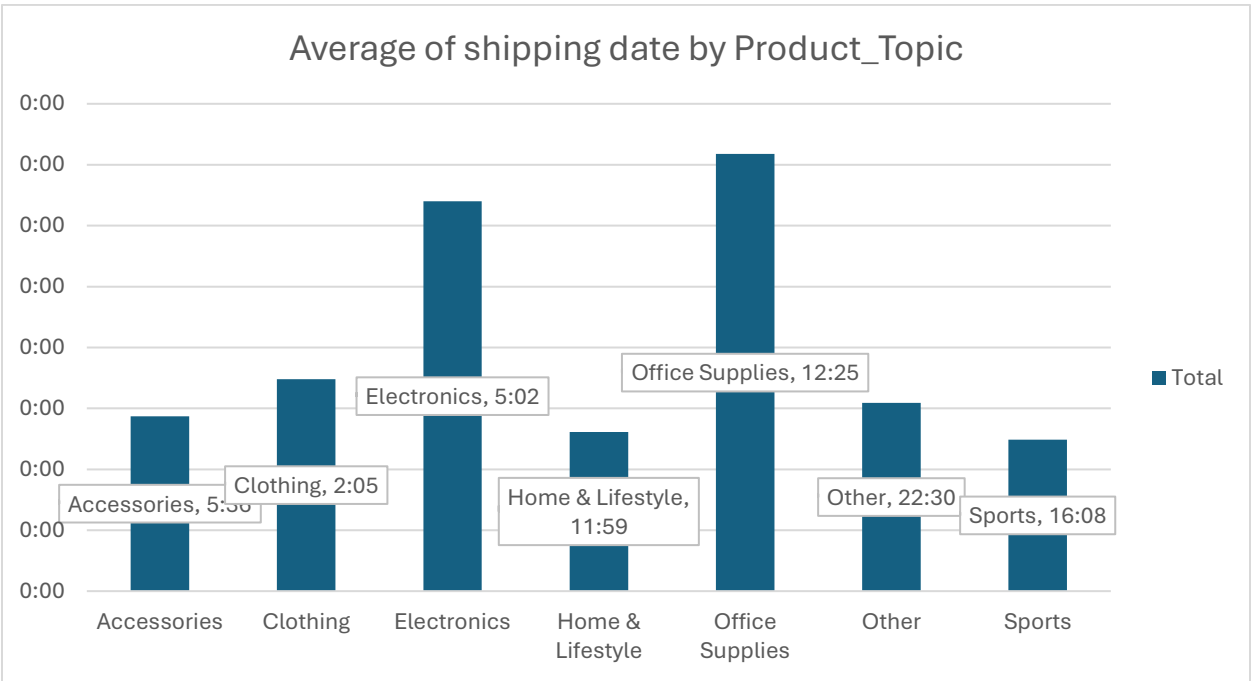
Electronics has the highest average product price (**\$607.64**), indicating this category consists of high-value items.

- **Home & Lifestyle** also shows a relatively high average (**\$298.67**), likely due to bigger household items.
- **Office Supplies** has the lowest average price (**\$25.11**), as expected for low-cost items like pens or notebooks.
- **Sports, Clothing, and Accessories** fall into the mid-range price levels.

The **overall average product price is \$154.22**, reflecting a mix of high-value and low-value categories.

Analytics for Decision Making

Product_Topic	Average of shipping date (DateOrders)
Accessories	5:36
Clothing	2:05
Electronics	5:02
Home & Lifestyle	11:59
Office Supplies	12:25
Other	22:30
Sports	16:08
Grand Total	11:23:42 PM



To measure delivery speed, calculate the time between order date and shipping date and average it by product topic. The office supplies took average time of 12:25 to deliver the orders.

Conclusion:

Topic Performance Analysis

The Topic Grouping analysis shows clear variation in performance across product categories. **Electronics** and **Sports** products contribute the highest total sales, driven by high-demand items

Analytics for Decision Making

such as laptops, cameras, sports shoes, and branded apparel. These categories also show strong profit margins, indicating they are key drivers of overall profitability.

Office Supplies and **Clothing** contribute moderate sales but display stable demand with consistent order patterns.

In contrast, **Home & Lifestyle** and **Accessories** categories generate lower revenue and slower delivery turnaround times, suggesting less operational efficiency or slower-moving inventory. The analysis highlights that focusing on improving supply chain processes for low-performing topics and maintaining stock levels for high-performing ones will support profitability and reduce inefficiencies.

Part 6 – Reporting and Recommendations

A. A short summary of the data preparation and filtering process.

The data preparation process transformed the raw dataset into a clean, structured, and enriched analytical foundation. Through cleaning, filtering, type correction, date normalization, and topic grouping, the dataset became suitable for advanced analyses such as forecasting, regression modeling, pivot summaries, and product-efficiency insights.

Dataset Overview:

The DataCo Smart Supply Chain dataset includes over **180,000 orders** with 50+ features covering customer, product, shipping, delivery, and financial data. Key variables include order dates, quantities, sales, delivery status, shipping times, regions, and late delivery risk.

Data Preparation & Filtering (Summary)

- Initial Data Load & Review
 - **Imported Europe_Data.xlsx** into the analysis environment.
 - Inspected column types, formats, and completeness.
 - Identified non-critical missing fields (Zipcode, Product Description).

Analytics for Decision Making

- Cleaning & Standardization
 - Verified no missing values in key analytical fields (sales, cost, dates, shipping days).
 - Converted text fields to consistent casing for keyword-based grouping.
 - Ensured numeric columns were correctly typed for aggregation and modeling.
- Date Normalization
 - Converted shipping date (DateOrders) into proper datetime format.
 - Created YearMonth field for monthly aggregation and forecasting.
- Feature Engineering
 - Built Product Topic using text-search logic (SEARCH/IF):
→ Grouped items into broad themes (Electronics, Sports & Outdoors, Apparel, etc.).
 - Created a temporary combined text field for category/topic classification.
 - Generated derived fields supporting pivot tables and analysis.
- Filtering & Validation
 - Confirmed dataset contained valid order records only.
 - Checked for duplicates or corrupted entries (none detected).
 - Ensured consistent date ranges (Jun 2015–Oct 2015).
- Aggregation for Analytics
 - Summarized sales, cost, and delivery metrics by:
 - Country
 - Category
 - Product Topic

Analytics for Decision Making

- Month
 - Prepared grouped data for forecasting, regression, and descriptive dashboards.

Month	Average of Days for shipping (real)	Sum of Sales per customer
9	3.492172824	\$2,089,123.80
8	3.489590141	\$1,946,104.80
7	3.463596119	\$1,925,360.22
10	3.481620718	\$1,842,757.99
6	3.519484083	\$1,324,853.17
11	3.51257485	\$355,735.26
12	3.326818675	\$162,640.51
1	3.614564831	\$102,391.28
5	3.727272727	\$20,231.28
Grand Total	3.487622383	\$9,769,198.32

1.2 Topic Grouping Preparation

A new worksheet was created to categorize each product into higher-level “product topics.”

Steps included:

- Copying **Product_Name** into a new sheet (Column AZ).
- Creating a blank column **Product_Topic** (Column BA).
- Preparing a reference keyword–topic table.
- Applying an IFS() formula using Excel SEARCH() to automatically assign items to topics such as *Electronics*, *Sports*, *Home & Lifestyle*, *Clothing*, *Office Supplies*, and

Analytics for Decision Making

Accessories.

Product Name	Count	Keyword	Product Topic
adidas Brazuca 2014 Official Match Ball	39	adidas	Sports
adidas Kids' F5 Messi FG Soccer Cleat	39	sweater	Clothing
adidas Men's F10 Messi TRX FG Soccer Cleat	68	Beverage Holder	Accessories
adidas Men's Germany Black Crest Away Tee	51	Bag	Accessories
adidas Youth Germany Black/Red Away Match Soc	68	Bowflex	Electronics
Baby sweater	82	Bridgestone	Electronics
Bag Boy Beverage Holder	57	shoe	Sports
Bag Boy M330 Push Cart	39	CDs	Office Supplies
Bowflex SelectTech 1090 Dumbbells	10	Cooler Bag	Accessories
Bridgestone e6 Straight Distance NFL Carolina	73	Laptop	Electronics
Bridgestone e6 Straight Distance NFL San Dieg	59	DVDs	Electronics
Bridgestone e6 Straight Distance NFL Tennesse	46	Diamondback	Home & Lifestyle
Brooks Women's Ghost 6 Running Shoe	40	Garmin	Electronics
Bushnell Pro X7 Jolt Slope Rangefinder	11	Glove	Clothing
CDs of rock	133	Computer	Electronics
Children's heaters	522	Sports Books	Office Supplies
Cleveland Golf Collegiate My Custom Wedge 588	38	Books	Office Supplies
Cleveland Golf Women's 588 RTX CB Satin Chrom	43	Golf	Sports
Clicgear 8.0 Shoe Brush	47	Armour	Sports
Clicgear Rovic Cooler Bag	47	Web Camera	Electronics
Columbia Men's PFG Anchor Tough T-Shirt	59	Watch	Electronics
Dell Laptop	350	LIJA	Clothing
Diamondback Boys' Insight 24 Performance Hybr	29	Titleist	Clothing
Diamondback Girls' Clarity 24 Hybrid Bike 201	28	Air Mattress	Home & Lifestyle
Diamondback Women's Serene Classic Comfort Bi	3793	Other	Other
DVDs	215	Shoe	Sports
Elevation Training Mask 2.0	39	Gun Fire	Sports
ENO Atlas Hammock Straps	67	Merrell	Sports
Field & Stream Sportsman 16 Gun Fire Safe	4815	Nike	Sports
Fitbit The One Wireless Activity & Sleep Trac	40	Fitness	Sports

< > ...

Product_Type Keyword

2.Descriptive Analytics

Pivot Table-Total Profit

Pivot Table_Average Delivery

... +

:

Ready Accessibility: Investigate

Data Filtering

To focus on the European supply chain:

- Filtered the dataset for **orders shipped to Europe** (EU countries only).
- Excluded orders with missing customer information or negative values.
- Filtered date range (if required) to include complete years for forecasting.

Analytics for Decision Making

B. Key insights from descriptive and diagnostic analyses.

Descriptive Analysis

- **Sales Concentration:** High-volume categories include **Sports & Outdoors** (Camping, Fishing, Cleats, Water Sports) and **Electronics** (Cameras, GPS).
- **Delivery Times:** Average ~3.5 days; trend indicates a gradual increase, suggesting emerging operational strain.
- **Market Distribution:** U.S. accounts for ~60% of total sales; Puerto Rico 40%. Delivery times are similar, indicating geography is not the key delay driver.
- **Category Behavior:** Bulky items (Water Sports, Camping Equipment, Fitness Machines) have longer shipping times; smaller items (Books, Accessories, Electronics) ship faster.

Details	Average (Mean)	Min	Max	Std Dev	Median
Days for shipping (real)	3.49	0	6	1.63	3.00
Order Profit Per Order	\$23.27	\$4,274.98	\$911.80	\$116.90	\$32.78
Sales per customer	\$194.40	\$8.47	\$1,939.99	\$145.88	\$167.99

C. Forecasting and Regression Results

1. Operational Forecasting

- **Trend:** Delivery efficiency is gradually declining.
- **Forecast:** Average shipping time expected to rise from 3.49 to ~3.60 days over the next three months.

Analytics for Decision Making

- **Implications:** Without intervention, late deliveries may increase, particularly for bulky categories.

2. Profitability Regression

- **Regression Findings:**
 - $R^2 = 1.7\%$; operational variables (shipping speed, discount rates, item quantities) do not significantly influence profit.
 - True profit drivers (~98% of variance) include **product category, margins, returns, logistics cost differences, and discount strategies.**
- **Recommendation:** Address **operational inefficiencies** and **margin/ pricing optimization** in parallel.

Regression Statistics	
Multiple R	0.132555098
R Square	0.017570854
Adjusted R Square	0.017492646
Standard Error	115.8745051
Observations	50252

The chart shown below shows

- **EE. UU.** significantly outperforms Puerto Rico in both **sales** and **cost volume** (reflecting higher order activity).
- **Delivery time** is nearly identical between the two regions (~3.5 days), meaning shipping performance is consistent regardless of location.
- Costs scale proportionally with sales, suggesting no major cost inefficiencies between countries.

Analytics for Decision Making

D. Actionable Recommendations for Europe Supply Chain

1. Reconfigure Fulfillment for Bulky Categories

- **Why:** Water Sports, Outdoor items, Fitness Equipment face highest delays and handling costs.
- **Actions:**
 - Create specialized handling zones for oversized SKUs.(identifies a product in inventory system)
 - Store bulky products in regional hubs closer to high-demand areas.
 - Partner with carriers optimized for oversized freight.
- **Impact:** Shorter shipping times, fewer late deliveries, lower logistics costs.

2 Rationalize Discounts & Margins for Low-Efficiency Categories

- **Why:** Apparel, novelty items, and certain Fitness products have high returns and discount use.
- **Actions:**
 - Implement tiered discount rules based on profitability.
 - Track return costs and penalize high-return SKUs.
 - Shift low-margin inventory into bundles or clearance cycles.
- **Impact:** Healthier margins, predictable profitability, reduced inventory aging.

3. Optimize Cross-Region Inventory Placement

- **Why:** Inventory positioning, not geography, drives delivery speed.
- **Actions:**
 - Use demand forecasts to redistribute stock across European fulfillment centers.
 - Position fast-moving Electronics & Outdoor SKUs near urban clusters.
 - Consolidate underperforming categories into fewer warehouses.
- **Impact:** Faster delivery, fewer stock-outs, improved warehouse utilization, leaner inventory footprint.

Analytics for Decision Making

Conclusion

The European supply chain network demonstrates strong overall revenue performance; however, underlying analytics reveal emerging operational and profitability challenges that require proactive intervention. Forecasting results indicate a rising trend in delivery delays, signaling increasing pressure on fulfillment capacity as order volumes grow. At the same time, regression analysis confirms that declining profitability is driven primarily by product category margin structures rather than shipping efficiency, highlighting a structural pricing and cost imbalance across categories.

By simultaneously executing these two strategic imperatives improving efficiency and optimizing profitability the organization can achieve **sustainable profitability**, significantly **reduce inefficiencies**, and build a **more resilient and scalable European supply chain** capable of supporting future growth ambitions and effectively navigating market volatility.